

Hoang Le

hoangleq765@gmail.com | C++/C, Python, Verilog HDL

EXPERIENCE

Viettel Solutions – Solution Consulting Intern

Ho Chi Minh City, Vietnam

Python, statsmodels, PostgreSQL/TimescaleDB, FastAPI, Gemini API, ONNX

Jun 2026 – Aug 2026

- Built an AI agent prototype that forecasts System Marginal Price (SMP) in Vietnam's wholesale electricity market (48 half-hour cycles per day), as an independent counterpart to a commercial project for a power generation company; handed over to the team head as a documented, reproducible archive
- Researched SMP formation (merit-order dispatch, price cap, wet/dry hydro regimes, solar duck curve), evaluated exogenous drivers and prioritized them by modeling value, and built a dual-write ingestion pipeline on PostgreSQL/TimescaleDB holding 95,000+ half-hourly cycles (2021–2026)
- Designed a zero-robust rolling-origin backtest (WAPE, 120 target days across 2021–2026) and compared SARIMAX harmonic regression, persistence baselines and an LSTM; built a multi-horizon forecast router reaching 6.9% WAPE at the next cycle and ~24% at day-ahead (2026 regime), and showed via error decomposition (81% shape-driven) that level-correction approaches cannot close the day-ahead gap
- Engineered a bilingual (Vietnamese/English) LLM orchestration layer with Gemini tool-calling: the model only routes and narrates while deterministic Python computes every number, backed by per-forecast provenance records and an offline fallback
- Delivered a FastAPI dashboard and chat console (48-cycle SMP curve with P10–P90 uncertainty band) and a local wake-word Vietnamese voice interface (Silero VAD, sherpa-onnx; ~0.04 s CPU transcription)

HCMUT - Smart Home System (AIoT)

Ho Chi Minh City, Vietnam

Python, MQTT, JavaScript, HTML/CSS

Feb 2026 – May 2026

- Engineered an IoT-based Smart Home system on an AIoT kit to automate and monitor residential environments
- Developed Python sensor interfaces for real-time temperature, humidity and air-quality monitoring, with automated control of fan speed (by ambient temperature) and lighting
- Designed a security module featuring abnormal motion detection, fire alarm triggers, and a PIN-based digital door lock
- Built a web dashboard (HTML, CSS, JavaScript, Chart.js) for real-time power consumption estimation, connected over MQTT via WebSockets to an Eclipse Mosquitto broker for low-latency remote control

ESP32 IoT and Edge AI Applications

Ho Chi Minh City, Vietnam

C++, ESP32, TinyML

Oct 2025 – Dec 2025

- Programmed an ESP32 in C++ for multi-tasking control of an LED and humidity-responsive NeoPixels, with temperature/humidity sensing shown on an LCD
- Configured the ESP32 to operate in Access Point mode, hosting a localized web server for direct user interaction and device control
- Deployed a TinyML model onto the edge device, conducting rigorous performance and accuracy evaluations for embedded machine learning capabilities

British Council

Ho Chi Minh City, Vietnam

Teaching Assistant

Jan 2025 – May 2026

- Collaborated continuously with native-speaking educators in a strict 100% English-speaking professional environment to deliver high-quality ESL instruction and manage classroom dynamics
- Developed engaging supplementary teaching materials and co-facilitated lessons to enhance student comprehension and academic participation
- Authored detailed daily progress reports and personalized feedback for parents utilizing Microsoft Word, ensuring transparent communication regarding student development

EDUCATION

Ho Chi Minh City University of Technology

Bachelor of Engineering in Computer Engineering

Ho Chi Minh City, Vietnam

Expected Graduation: 2027

- **Relevant Coursework:** Digital Logic and VLSI: Digital Systems, Logic Design with HDL (+Lab), LSI Logic Design (+Lab), Introduction to SoC (+Lab), Embedded and Architecture: Microprocessors-Microcontrollers (+Lab), Computer Architecture (+Lab), Electronics: Electrical Electronic Circuits (+Lab), Computer Science: Data Structures and Algorithms (DSA), Mathematical Modeling

SKILLS, ACTIVITIES, AND INTERESTS

Languages: Fluent in English (IELTS: 7.0 - 2023) and Vietnamese

Technical Skills: Programming Languages: C/C++, Python, Verilog HDL; Data and AI: time-series forecasting (SARIMAX, rolling-origin backtesting), Pandas, NumPy, statsmodels, LLM tool-calling (Gemini API), ONNX speech processing; Backend and Data: FastAPI, PostgreSQL/TimescaleDB; Software and Tools: STM32CubeIDE, Xilinx Vivado, Proteus, MobaXterm, VMware; Operating Systems: Linux (Ubuntu), Windows; IoT and Web Technologies: MQTT, WebSockets, HTML/CSS/JavaScript, Chart.js; Hardware and Platforms: ESP32, AIoT Kits

Activities: Runner-up at “Sinh Viên Thế Hệ Mới” with Aloe vera food wrap research

Interests: Badminton (12 years), Soccer (12 years), Chess(1 year)